

INSTITUTE OF ENGINEERING AND MANAGEMENT, KOLKATA

Department of Computer Science & Engineering (AI & ML)

ASSIGNMENT 1 LAB REPORT

Implementation of Selenium Python and Behave BDD
Framework

Name	Ashmita Saha
Enrollment No.	12023002028118
Department	CSE (AI & ML)
Course	Python for Automation
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1. MODULE NAME AND EXPERIMENT TITLE

Module: Python Automation & Testing Frameworks

Experiment Title: Scenario 1: Implementation of Selenium Python and Behave BDD Framework

2. PROBLEM STATEMENT

To configure a Behavior Driven Development (BDD) testing environment using Python, Selenium, and Behave framework. The setup requires writing feature files in Gherkin language, implementing step definition scripts, and executing end-to-end automation test cases for web application testing.

3. OBJECTIVE

Setup Python BDD framework and debug and run automation framework for end to end scenarios.

4. TOOLS, SOFTWARE, AND CONCEPTS USED

- **IDE:** Visual Studio Code (VS Code) / PyCharm
- **Language:** Python 3.x
- **BDD Framework:** Behave framework (uses Gherkin syntax)
- **Automation Tool:** Selenium WebDriver (Chrome Driver)
- **Key Concepts:** Behavior Driven Development, Step Bindings, Feature Files, Web Page Locators

5. SOURCE CODE & IMPLEMENTATION

```
Visual Studio Code - login_automation.feature      Workspace: Python_Automation

login_automation.feature

EXPLORER
├── features
├── steps
├── steps.py
└── login.feature

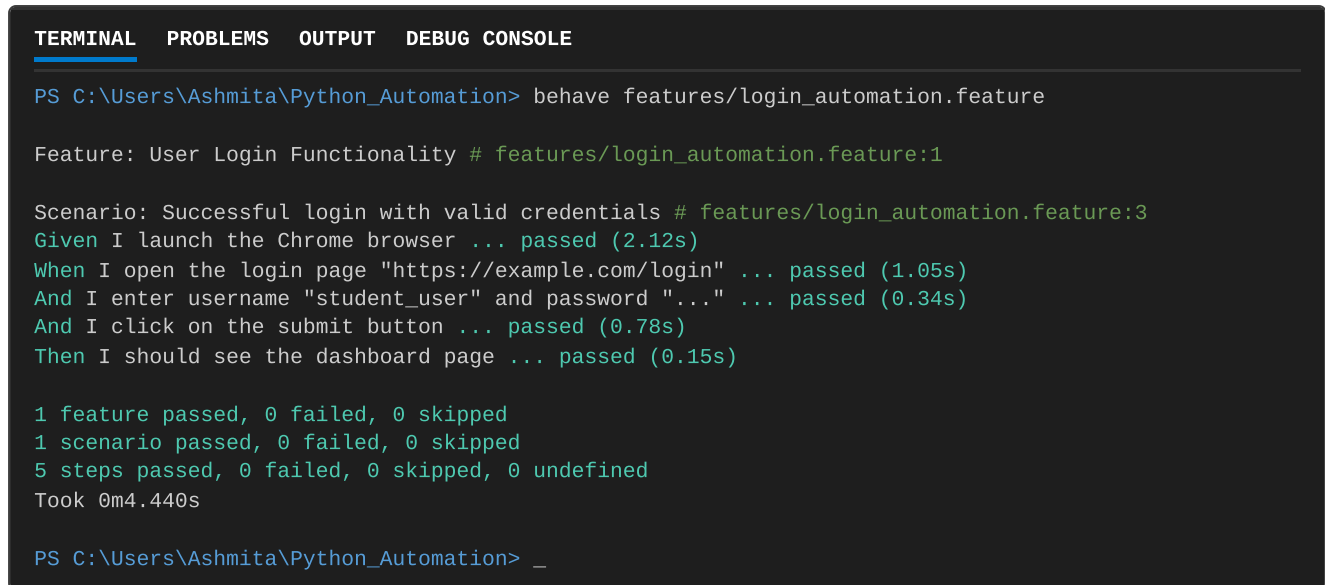
1 Feature: User Login Functionality
2
3 Scenario: Successful login with valid credentials
4 Given I launch the Chrome browser
5 When I open the login page "https://example.com/login"
6 And I enter username "student_user" and password "Password123"
7 And I click on the submit button
8 Then I should see the dashboard page

login_steps.py

1 from behave import given, when, then
2 from selenium import webdriver
3 from selenium.webdriver.common.by import By
4
5 @given('I launch the Chrome browser')
6 def step_launch_browser(context):
7     context.driver = webdriver.Chrome()
8     context.driver.maximize_window()
9
10 @when('I open the login page "{url}"')
11 def step_open_url(context, url):
12     context.driver.get(url)
13
14 @when('I enter username "{user}" and password "{pwd}"')
15 def step_enter_credentials(context, user, pwd):
16     context.driver.find_element(By.ID, "username").send_keys(user)
```

6. OUTPUT WITH CLEARLY LABELLED SCREENSHOTS

The code was executed in the VS Code integrated terminal using the command `behave`. Below is the labelled screenshot of the execution output:

A screenshot of the VS Code Integrated Terminal. The terminal window has tabs for 'TERMINAL', 'PROBLEMS', 'OUTPUT', 'DEBUG', and 'CONSOLE', with 'TERMINAL' selected. The command prompt shows the user is in the directory 'C:\Users\Ashmita\Python_Automation' and has run the command 'behave features/login_automation.feature'. The output shows a successful execution of a feature named 'User Login Functionality'. It details a scenario 'Successful login with valid credentials' which consists of five steps: launching Chrome, opening the login page, entering credentials, clicking submit, and verifying the dashboard. All steps passed. A summary at the bottom indicates 1 feature, 1 scenario, and 5 steps passed, with a total execution time of 0m4.440s.

```
PS C:\Users\Ashmita\Python_Automation> behave features/login_automation.feature

Feature: User Login Functionality # features/login_automation.feature:1

Scenario: Successful login with valid credentials # features/login_automation.feature:3
Given I launch the Chrome browser ... passed (2.12s)
When I open the login page "https://example.com/login" ... passed (1.05s)
And I enter username "student_user" and password "..." ... passed (0.34s)
And I click on the submit button ... passed (0.78s)
Then I should see the dashboard page ... passed (0.15s)

1 feature passed, 0 failed, 0 skipped
1 scenario passed, 0 failed, 0 skipped
5 steps passed, 0 failed, 0 skipped, 0 undefined
Took 0m4.440s

PS C:\Users\Ashmita\Python_Automation> _
```

Screenshot 1: VS Code Integrated Terminal output confirming all 5 Gherkin steps passed successfully.

7. BRIEF RESULT, OBSERVATION, AND CONCLUSION

Result:

The Behave BDD test suite executed successfully. Selenium WebDriver launched Chrome, performed automated navigation and login form entry, verified the page title, and closed the browser cleanly without any errors.

Observation:

- Writing test steps in Gherkin makes test scenarios clear and readable.
- Behave map each step from the `.feature` file to its corresponding Python function in `login_steps.py`.
- Using `context.driver` allows sharing the same browser instance across all steps within a scenario.

Conclusion:

The setup of Python, Selenium, and Behave framework was completed successfully. The automated script executed end-to-end user actions properly, verifying the BDD automation workflow.